Process Characterization Plan

A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) — PCP-007

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; ALCOA+, attributable, legible, contemporaneous, original, accurate and complete, consistent, enduring, available; AMV, analytical method validation; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; ELISA, enzyme linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; IgG1, immunoglobulin G isotype; KPP, key process parameter; MSAT, Manufacturing Science and Technology; NOR, normal operating range; OD, optical density; PAR, proven acceptable range; PD, Process Development; PPQ, process performance qualification; qPCR, quantitative polymerase chain reaction; QA, Quality Assurance; RSM, response surface methodology; SDM, scale-down model; SEC, size exclusion chromatography; SOP, standard operating procedure; UV, ultraviolet; WC-CPP, well-controlled critical process parameter.

1 Purpose and scope

The A-Mab drug substance process is being transferred from Novacyte Biologics — Cambridge, MA (Development) to Novacyte Biologics — Grafton, WI (Commercial DS) under the transfer plan PTP-001. This plan describes the Stage 1 characterization of the cation exchange chromatography step, which is Step 7 of the drug substance train and its principal polishing operation. The studies will be executed on a qualified scale-down model of the commercial column, in support of a process that operates at a 15,000 L production bioreactor scale at the receiving site. Their purpose is to measure how far the aggregate content, the host cell protein content and the yield of the eluate pool move as each parameter is varied across the ranges in Table 6.1, to define the region over which the pool meets its criteria, and to justify the ranges and the controls that will be applied in commercial manufacture (U.S. Food and Drug Administration 2011).

The scope of this plan covers the 5 process parameters of the step, the size variant and impurity attributes the step clears, and the step yield. 4 parameters will be studied in a multivariate design and 1 will be assessed one at a time, on the assignment made in RA-001. The feed to the step is the neutralized pool of the low pH viral inactivation step, and its attributes are treated as inputs to the study rather than as factors of it.

Three activities lie outside this plan. Resin life cycle, sanitization and repacking are qualified separately under SOP-2008, and the studies here will be run on resin inside that qualified life. Validation of the analytical methods is covered by the method validation reports in Table 3.2, and this plan uses those methods as validated. Confirmation of the ranges at commercial scale belongs to Stage 2 and to the process performance qualification batches, which follow the transfer.

No viral clearance claim is made for the cation exchange step. Clearance of enveloped and small viruses is claimed at the low pH inactivation step, the anion exchange step and the virus filtration step, and the cumulative claim is consolidated in PCMR-001 (International Council for Harmonisation 2023a). Cation exchange contributes clearance of host cell protein, DNA and leached Protein A, and those contributions are within scope.

2 Objectives

The study has five objectives.

1. To quantify the relationship between each studied parameter and each response of the step, over ranges wider than the ranges in which the step will be operated.
2. To define the multivariate region over which the fitted models predict that the aggregate and the host cell protein content of the eluate pool both meet the criteria in Table 7.1.
3. To establish a proven acceptable range for each parameter against the aggregate and the host cell protein content of the pool, both with the rest of the step at target and with the rest of the step varying within its normal operating ranges.
4. To classify each parameter under SOP-4001, from the size of its demonstrated effect and from the capability with which the parameter is controlled.
5. To state the ranges, in-process controls and monitoring this step contributes to the control strategy, and to justify them for Stage 2.

The report PCR-007 will present the outcome against each of these objectives.

3 Related documents

Table 3.1 lists the documents this plan depends on and the document it delivers. RA-001 set the scope of the study, this plan executes that scope, and PCR-007 will report the outcome. The controlled procedures and method validations that govern the work are listed in Table 3.2.

Table 3.1: Related documents in the characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-007	Process Characterization Report (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2: Controlled procedures and analytical method validations that govern this study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation

Reference	Title	Type
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The cation exchange step is operated in bind and elute mode and is the principal aggregate polishing operation of the purification train. Below its isoelectric point the antibody carries a net positive charge and binds the sulfonate ligand of the resin. It is eluted by raising the pH of the elution buffer, which weakens the electrostatic attraction. Aggregate presents more positive surface to the ligand than monomer does and is therefore retained more strongly, so it elutes later than monomer. The clearance the step achieves depends on how well the two species are resolved and on the point at which pool collection is stopped.

Host cell protein, DNA and leached Protein A are cleared because they carry a net negative charge at the load pH and are therefore not retained by the sulfonate ligand. Most host cell proteins are acidic at the operating pH, so they leave in the load flow through, and the weakly bound remainder is displaced during the wash. DNA carries a strong negative charge and is repelled by the ligand, so it passes through in the load and its clearance is large and insensitive to the operating parameters. Leached Protein A and its fragments are acidic as well and do not bind under load conditions, and the fraction that is complexed with the antibody is released when the antibody binds the ligand. The step therefore clears leached Protein A by a modest factor that is a property of the chemistry rather than of any parameter studied here.

Yield is lost at the two edges of the pool. Monomer that elutes beyond the stop collect point is left in the tail, and monomer that co-elutes with the aggregate that is cut away is lost with it.

The step has been operated on Novacyte's humanized IgG1 platform for three previous products (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical and engineering manufacture. The resin chemistry, the buffer system and the mode of operation are unchanged for A-Mab. This experience establishes the mechanisms above and identifies the parameters worth studying, so the studies in this plan are intended to bound and confirm platform behaviour rather than to discover it. Prior knowledge is not carried across as a numerical claim. Every range and every criterion reported in PCR-007 will rest on data generated under this plan.

Four expectations follow from that experience, and the multivariate design is built to test them.

- A higher protein load occupies more of the sulfonate ligand, so at the top of the studied range the column approaches its dynamic binding capacity. As occupancy rises the monomer and the aggregate peak both broaden and overlap further, and weakly bound host cell protein finds fewer free ligand sites from which the wash would otherwise displace it. Rising ligand occupancy is therefore expected to move aggregate clearance, host cell protein clearance and yield in the same direction, all three falling as load rises.

- Load and wash conductivity sets the ionic strength that shields the charge on the ligand and on the bound species. A higher conductivity is expected to strip more of the weakly bound host cell protein during the wash, while the antibody, which binds more strongly, stays bound. Its effect should fall on impurity clearance rather than on the aggregate separation, which happens during elution.
- Elution buffer pH sets the net charge on the antibody at the moment of elution. A higher pH within the studied range weakens binding and brings monomer and aggregate off the column earlier and closer together, so it is expected to reduce the resolution on which aggregate clearance depends.
- Monomer elutes ahead of aggregate, so the material still leaving the column on the descending edge of the peak is progressively richer in aggregate. Stopping collection later therefore recovers more monomer from the tail and carries the leading part of the aggregate peak into the pool, and stopping earlier leaves both behind.

Elution flow rate sets the residence time of the antibody in the pores of the resin during elution, and a longer residence time lets the antibody equilibrate more completely with the mobile phase and so sharpens the elution peak. Over the range in Table 6.1, and at the particle size of the platform resin, the change in peak sharpness is expected to be too small to move the aggregate content of the pool.

4.2 Quality attributes in scope

The cation exchange step forms no quality attribute of its own. Aggregate is formed in the production bioreactor and is added to during the low pH hold, host cell protein and DNA enter with the harvested cell culture fluid, and leached Protein A is released from the capture resin. This step removes them. Table 4.1 lists the four attributes in scope with their drug substance acceptance criteria and the criticality assigned to each. Criticality follows the continuum of ICH Q9 rather than a binary split (International Council for Harmonisation 2023b), and it was assigned with Tool #1 of the case study, which combines the impact of the attribute on safety and efficacy with the uncertainty in that impact (CMC Biotech Working Group 2009). Attributes of high and very high criticality are called critical.

Table 4.1: Quality attributes cleared by the cation exchange step, with drug substance acceptance criteria and assigned criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36

CQA	Category	Acceptance	Criticality	Tool #1
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

Two of the four attributes will be measured as responses of the designed experiments. Aggregate is the attribute the step exists to control and carries the highest criticality of the four. Host cell protein is cleared here and again at the anion exchange step, so this step must hand on a pool that the remaining train can bring to the drug substance criterion. Residual DNA and leached Protein A will be measured across the step so that their clearance is confirmed, and neither is treated as a response of the designed experiments, because DNA is repelled by the sulfonate ligand at every conductivity and pH in Table 6.1 and leached Protein A stays unbound over the same range. PCR-007 will report the clearance observed for both, and will claim a parameter dependence for neither unless the data show one.

Step yield is a process performance attribute rather than a quality attribute. It is measured on every run because the monomer that is cut away with the aggregate is lost from the pool, so the aggregate clearance and the yield of a setting have to be read together. It has no acceptance criterion in Table 4.1.

4.3 Risk-based prioritization of parameters

RA-001 set the scope of this study. All process parameters of the cation exchange step were carried into a risk ranking and filtering exercise, in which each parameter was given one ranking for its potential impact on a quality attribute and a second for its potential to interact with other parameters, and the two rankings were combined into a study assignment (CMC Biotech Working Group 2009). Where no data or rationale were available to make an assessment, the parameter was ranked at the highest level.

The assignment produced three groups. 4 parameters were assigned to the multivariate design. Protein load, elution buffer pH and elution stop collect each move the aggregate peak relative to the point at which collection stops, and protein load acts with the load and wash conductivity to set how much weakly bound host cell protein survives the wash, so the effects of the four on the pool cannot be assumed to add. Elution flow rate was assigned to a univariate assessment, since it changes the residence time of the antibody in the pores of the resin rather than the strength with which the antibody binds the ligand, and its expected effect on the separation is small. Parameters that are fixed by equipment design or held to platform specification, such as bed height, column packing quality and buffer composition, were not assigned a new study. Their ranges are supported by the platform knowledge space and by the operation and life cycle procedures in Table 3.2.

The ranges in Table 6.1 are wider than the normal operating ranges in every case. Each was set from the control capability of the commercial equipment and from the range over which the platform has been operated, so that the study bounds the step beyond the region in which it will run and supports later movement within the region it defines (International Council for Harmonisation 2009). Ranges were not widened past the point at which the step would fail for a reason that is already understood. The elution buffer pH range stops short of the pH at which the antibody would no longer bind reproducibly, because a run at such a pH would confirm the mechanism and would tell the study nothing about the operating region.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be run on a laboratory column packed with the same resin type and to the same bed height as the commercial column, and operated at the same linear velocities. Protein load will be set in grams of antibody per litre of resin, and the equilibration, wash, elution and strip volumes will be set in column volumes, so that every parameter of the study is expressed at its commercial equivalent. Packing quality will be verified before each campaign by plate count and peak asymmetry, and a column that fails either test will be repacked before use. The model and its operation are described in SOP-1001 and SOP-2010.

The model will be qualified against commercial scale data before any characterization run is executed. Qualification will compare the model and the commercial step at the set-point condition of Table 6.1 on the attributes that enter and leave the step, which are step yield, pool aggregate, pool host cell protein, residual DNA and leached Protein A. The comparison will be made from replicate runs at small scale against the available at-scale batches, and the model will be accepted where no statistically significant difference is found on any of those attributes. Where a difference is found, its cause should be established and the model modified before characterization proceeds, as SOP-1001 requires.

Reproducibility of the qualified model will be re-established inside each designed experiment. The centre point runs of both designs are executed at the set-point of every factor, so their spread measures the reproducibility of the model under study conditions and provides the pure error term against which lack of fit is tested (see §5.5). It is the only estimate of experimental error the analysis uses.

The claims this study makes are bounded by the model. A scale-down column reproduces the chromatography of the commercial column, and it does not reproduce the hold times, the pool volumes or the mixing of the commercial suite. Conclusions drawn here apply to the commercial step through the qualification above, and they will be confirmed at scale during Stage 2.

5.2 Operation

The step receives the neutralized pool of the low pH viral inactivation step. The pool is adjusted to the load conductivity given in Table 6.1 and is loaded onto the equilibrated column at the target protein load. The column is then washed at the same conductivity as the load, so one buffer sets the ionic strength for both operations and the two cannot be varied independently. Elution is a step change to the elution buffer at the pH under study. The pool is collected between a threshold on the ascending edge of the elution peak and the stop collect threshold on the descending edge, both read as ultraviolet absorbance relative to the peak maximum. After collection the column is stripped, sanitized and re-equilibrated under SOP-2008.

Buffer preparation, column packing and resin storage are held to platform specification and are governed by the procedures in Table 3.2. They are identity and quality controlled inputs to this study rather than factors of it. Chromatography is performed under ambient temperature control and temperature is not varied, because the platform operates the step at controlled room temperature at both sites and the transfer introduces no difference in that control.

5.3 Analytical methods

Every response and every attribute in scope will be measured by a validated method. Table 5.1 lists the method validation reports that apply. Pool aggregate will be measured by size exclusion chromatography under AMV-3011 and reported as the percentage of high molecular weight species. Host cell protein will be measured by enzyme linked immunosorbent assay under AMV-3012 and reported per milligram of antibody, so that the reported value does not move with the concentration of the pool. Residual DNA will be measured by quantitative polymerase chain reaction under AMV-3014, and leached Protein A by enzyme linked immunosorbent assay under AMV-3016. Step yield will be calculated from the antibody mass in the pool and in the load, each determined by ultraviolet absorbance against the extinction coefficient of A-Mab.

Table 5.1: Validated analytical methods used in the study.

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

Assay variability enters the analysis at two points and is handled at both. It is part of the residual variance of each fitted model, and it is part of the centre point spread that estimates pure error, so a response measured by a less precise assay will carry a wider prediction interval and a less sensitive lack of fit test. Where a predicted value sits within assay variability of its criterion, PCR-007 should attribute the result to the assay and should not claim an effect. Host cell protein is expected to be the least reproducible of the three responses, and the centre point runs will establish whether that expectation holds.

5.4 Sampling plan

Each run will be sampled at four points. The load will be sampled after conditioning, so that the aggregate, host cell protein, DNA and leached Protein A levels entering the step are known for every run. The flow through and the wash will each be sampled as a single pooled fraction, which allows the host cell protein mass balance across the step to be closed. The eluate pool will be sampled after collection and mixing are complete, and it carries every response of the study. A strip fraction will be sampled and retained, and will be tested only where a run fails to close its mass balance.

Replication is concentrated at the centre point. The screening design carries 3 centre point runs and the response surface design carries 4, each executed at the set-point of every factor. Samples will be assigned to analytical runs in an order that does not follow the design order, so that drift in an assay cannot align with a factor. Retained samples will be held under the conditions and for the period stated in SOP-2010, so that a re-test is possible where a result is questioned.

5.5 Statistical methods

All factors will be analysed in coded units. Each factor is coded so that its set-point is 0 and the edges of its studied range are -1 and +1, which makes the estimated effects comparable across factors carrying different units and makes the intercept the predicted response at the centre of the design. The letter codes are given in Table 6.2 and are used for the terms of both models.

The screening design will be analysed by ordinary least squares with all main effects and all two factor interactions. An effect will be reported as twice the coded coefficient, which is the change in the response across the full studied range of that factor. Terms will be judged significant at $p < 0.05$. The purpose of the screening analysis is to identify which factors and which interactions change each response, and by how much across the studied range. It is not the predictive model of the step, since a model without quadratic terms cannot describe curvature, and the response surface model is the model from which the operating region, the proven acceptable ranges and the capability estimate will all be derived.

The response surface design will be analysed as a full quadratic model in its 4 factors, with main effects, two factor interactions and pure quadratic terms. Model adequacy will be judged on four things together. The overall F test of the model should be significant at $p < 0.05$. The coefficient of determination, its adjustment for the number of terms in the model, and the predicted coefficient of determination computed from the prediction error sum of squares should lie close to one another, because a predicted value well below the adjusted value indicates a model that describes the runs it was built on and predicts poorly. The lack of fit mean square will be tested against the pure error mean square from the centre point replicates, and a significant lack of fit means the model form is inadequate for that response. Residuals will be examined against the fitted values and on a normal quantile plot, and any run that carries the fit will be identified.

A factor with no significant term in either model will be reported as having no demonstrated effect on that response. Such a statement is bounded by the range studied and by the power the design carries, and PCR-007 will state it in those terms rather than as an absence of effect in general.

Capability at commercial scale will be estimated by Monte Carlo simulation. 2,000 batches will be simulated with every parameter drawn within its normal operating range, the fitted models will be evaluated at each draw, and a one sided capability index will be computed for each attribute in Table 7.1 against its criterion. No minimum capability index is set as an acceptance criterion in this plan. The decision rule is the one stated in §7, and the capability estimate will be reported alongside it, with the margin between the predicted distribution and the limit, as supporting evidence.

The proven acceptable range analysis is described in §8. Both of its analyses use the response surface model, and its Monte Carlo component uses a fixed random seed, so that a repeat of the analysis on the recorded data returns the same ranges.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 gives the parameters of the step with their set-points, the range each will be studied over, the normal operating range and the study assigned to it. The table carries no classification, because classification is an outcome of the study and will be reported in PCR-007 under SOP-4001.

Table 6.1: Parameters of the cation exchange step, the range each will be studied over, the normal operating range and the assigned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	25	10–40	18–32	multivariate
Load/Wash conductivity	mS/cm	5	3–7	4–6	multivariate
Elution buffer pH	pH	6	5.8–6.2	5.85–6.15	multivariate
Elution stop collect	OD desc.	1	0.5–1.5	0.7–1.3	multivariate
Elution flow rate	cm/hr	200	100–300	150–250	univariate

Table 6.2: Coded factor legend for the screening and response-surface designs.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

The 4 multivariate parameters enter both designs as the factors of Table 6.2. The letter codes are used for the model terms in PCR-007 and for the design matrices in Appendix A and Appendix B.

6.2 Screening design

The screening study is a two level full factorial in the 4 multivariate factors, augmented with 3 centre point runs, giving 19 runs in total. A full factorial was selected rather than a fraction, because the factor count is small and because protein load is expected to interact with each of the other three factors, through the peak broadening that rising ligand occupancy produces. A resolution IV fraction would alias those interactions with each other and would leave the analysis unable to say which of them is active. The design as planned estimates every main effect and every two factor interaction independently.

Runs will be executed in a randomized order, with the centre point runs distributed through the campaign rather than grouped, so that drift in the model or in the feed material appears in the centre point spread rather than in a factor effect. The planned coded design is given in Appendix A. All runs will be performed on a single conditioned feed lot where the available material allows it. Where more than one lot is required, the lot will be recorded with the run and the centre point runs will be balanced across lots.

6.3 Response-surface design

The response surface study is a face centred central composite design in the 4 factors of the screening study, with 16 factorial runs, 8 axial runs and 4 centre point runs, giving 28 runs in total. The axial runs are placed on the faces of the cube rather than beyond them, so that no run is executed outside the studied range of any factor. The studied ranges are already wider than the normal operating ranges, and a rotatable design would require settings outside them for which the step has no prior support.

The response surface design adds the quadratic terms that the screening design cannot estimate. Curvature is expected on elution stop collect and on elution buffer pH. Both change where the pool is cut relative to the aggregate peak, and because that peak has a maximum, equal steps in the cut point add unequal amounts of aggregate to the pool. The planned coded design is given in Appendix B.

The two designs are executed in sequence and are analysed separately. Should the screening analysis identify an active factor that is not carried into the response surface design, the design should be augmented before the operating region is declared, and the augmentation should be documented as a change to this plan under §13.

6.4 Univariate assessment

Elution flow rate will be assessed one factor at a time. Runs will be executed at the low edge, the set-point and the high edge of the range given in Table 6.1, with every other parameter held at its set-point, and the responses of §5.3 will be measured on each. The assessment will establish whether a longer residence time sharpens the elution peak enough to change the aggregate content or the yield of the pool over the studied range, and it will support the range within which flow rate is controlled in manufacture.

A univariate assessment cannot determine whether flow rate interacts with the other parameters. It is accepted on that basis here because flow rate changes the residence time of the antibody in the pores of the resin and with it the sharpness of the elution peak, and it changes neither the net charge on the antibody nor the ionic strength that sets how strongly the antibody is held, so an interaction with load, conductivity, elution pH or the collection window is not expected. Should the assessment show an effect on any response comparable in size with the effects estimated in the screening study, flow rate should be carried into a multivariate study before the operating region is declared.

7 Acceptance and decision criteria

This section states the criteria that will be applied, before any data exist. Four decisions will be made under this plan, and each has its own rule.

The scale-down model is accepted where the qualification of §5.1 finds no statistically significant difference between the model and the commercial step on any of the compared attributes. A model that fails that comparison will not be used for characterization, and the difference should be investigated and resolved first.

A model of a response is accepted as adequate where its overall F test is significant at $p < 0.05$ and its lack of fit is not significant against pure error at the same level. A response whose model is not adequate will be reported as characterized by its data rather than by a model, and no design space and no proven acceptable range will be derived from it. Where lack of fit is significant, PCR-007 should state whether the cause is a missing term or an outlying run, and should establish that from the residuals rather than from the summary statistics alone.

The quality criteria the pool is judged against are given in Table 7.1.

Table 7.1: Acceptance criteria applied to the responses of the study. The in-process criterion is the level the eluate pool of this step must meet.

Response	Quality attribute	Criticality	Drug-substance criterion	In-process criterion for this step
Aggregates (HMW, %)	Aggregates (HMW)	H	≤ 5 % HMW (SEC)	≤ 1 % HMW (SEC)
Pool HCP (ng/mg)	Host Cell Protein (HCP)	M-H	≤ 100 ng/mg	≤ 193 ng/mg

The drug substance criterion is the level the batch must meet at release. The in-process criterion is the level the eluate pool of this step must meet. For aggregate the in-process criterion is the tighter of the two, because no step downstream of cation exchange dissociates high molecular weight species, so the aggregate that leaves this step reaches the drug substance as aggregate. For host cell protein the in-process criterion is the wider of the two, because the anion exchange step binds host cell protein while the antibody passes through and removes a further part of it before formulation. Each in-process criterion was calculated back from the drug substance criterion through the clearance the downstream steps deliver when the process is operated at its set-points, and each was then divided by a safety factor, 5 for aggregate and 3.2 for host cell protein. The safety factor keeps the pool inside the drug substance criterion where a downstream step delivers less clearance than it delivers at its set-point.

The operating region will be declared where the fitted models predict that both attributes in Table 7.1 meet their in-process criteria, evaluated over the studied ranges of the 4 multivariate factors. The declared region will lie inside the ranges studied, and the normal operating ranges will lie inside the declared region. PCR-007 will state the margin between the two and will identify the attribute and the parameter that bound the region most tightly. Where the models predict that a criterion is met across the whole studied range of a parameter, that will be reported as the finding, and the range will not be narrowed to manufacture a margin the data do not require.

Step yield has no quality criterion and does not bound the operating region. It will be reported as a process performance attribute across the region, and a setting that meets every quality criterion at a cost in yield will be described as such rather than excluded.

A parameter will be classified under SOP-4001 from two things together, the size of the effect the study demonstrates on a quality attribute and the risk that the parameter moves outside the declared region in manufacture (CMC Biotech Working Group 2009). A parameter linked to a quality attribute and held comfortably inside the region by the commercial control system is a well-controlled critical process parameter. A parameter linked to a quality attribute whose control is narrow relative to the region is a critical process parameter. A parameter that changes yield or another performance attribute without changing a quality attribute is a key process parameter. A parameter that changes neither over the range studied is a general process parameter. The classifications themselves are outcomes and are reported in PCR-007.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range will be determined for each parameter of the multivariate design against the aggregate and the host cell protein content of the pool. The criterion applied is the in-process criterion for that attribute in Table 7.1, so a proven acceptable range in this study is the range of a parameter over which the fitted model predicts that the eluate pool meets the level the step must hand on.

Two analyses will be run for every pair of parameter and attribute, and both will be reported. The first evaluates the response surface model across the studied range of the parameter with the other three factors held at the centre of the design, which for every parameter of this step is its set-point (Table 6.1). It answers what one parameter can do while the rest of the step is at target. The second propagates the normal operating ranges of the other three factors. At each of 81 settings across the studied range of the parameter, 2,000 draws will be taken with the other factors distributed within their normal operating ranges, the model will be evaluated at every draw, and a setting will be accepted only where the predicted values meet the criterion across those draws. The second analysis is the one that bounds routine operation. It will in general return the narrower of the two ranges, because a setting that is acceptable with the rest of the step at target need not be acceptable when the rest of the step sits anywhere inside its normal operating ranges.

The reported proven acceptable range is the contiguous interval containing the set-point over which the criterion is met. Where the acceptable settings are not contiguous, the contiguous part containing the set-point will be reported and PCR-007 will state that this has been done. Where the criterion is met across the whole studied range, the proven acceptable range is the studied range, and the claim is bounded by that range and no wider.

Each of the two attributes will be reported with a plot of the predicted response against the parameter carrying the largest main effect on it, with the acceptable region shaded, the set-point marked and the normal operating range marked. The Monte Carlo component uses the fixed seed 20240724, so a repeat of the analysis on the recorded data returns the same ranges.

A proven acceptable range is a statement about one parameter while the others are held at defined settings. The operating region is a statement about the 4 factors together, and a combination of settings that are each inside their proven acceptable range may still fall outside the region where an interaction is active. Both will be reported, and the control strategy will be built on the region.

9 Data management and integrity

Data generated under this plan will be recorded and retained under the ALCOA+ principles, so that every result is attributable, legible, contemporaneous, original and accurate, and so that the record is complete, consistent, enduring and available. Run conditions and observations will be recorded in the electronic laboratory notebook at the time of execution. Chromatograms and their integration will be retained in the chromatography data system under audit trail. Analytical results will be transferred from the validated instrument systems into the study dataset without manual transcription wherever the systems support it, and any manual entry will be verified by a second person.

The analysis dataset, the analysis scripts and the fitted models will be retained with the report, so that every number in PCR-007 can be traced to the run that produced it. A recorded value may be changed after review only under the correction procedure, with the original value, the reason and the author retained. Deviations from this plan will be recorded as they occur and will be reported in PCR-007 with their investigation, their root cause and their impact on the conclusions, whether or not a conclusion changes as a result.

10 Roles and responsibilities

Process Development and MSAT own this plan. They design the studies, execute the runs on the scale-down model, analyse the data and author PCR-007. The other functions in Table 10.1 review and approve within their own competence, and no function approves its own work.

Table 10.1: Functions and their responsibilities under this plan.

Function	Responsibility
Process Development / MSAT	Owns the plan, designs and executes the studies, analyses the data, authors PCR-007
Analytical Development	Runs the validated methods, releases results, advises on assay variability
Manufacturing Science (receiving site)	Confirms that the proposed ranges are ranges the commercial equipment can hold
Statistics	Reviews the designs before execution and the fitted models after it
Quality Assurance	Approves the plan and the report, reviews deviations raised under the plan

A role may be held by more than one person, and one person may hold more than one role, provided that the person who reviews an analysis is not the person who performed it. The names of the individuals holding each role at execution are recorded in the study file rather than in this plan, so that a change of personnel does not require a revision of the plan.

11 Deliverables and schedule

The deliverable of this plan is PCR-007. It will report the execution of every study described here, the models fitted to the data, the operating region, the proven acceptable ranges, the capability estimate and the classification of each parameter. It will also carry the design matrices as executed, including any run that was repeated, and every deviation recorded during the campaign. PCR-007 rolls up into PCMR-001, which consolidates the characterization of the whole train.

The work runs in five phases. The scale-down model is qualified first, and no characterization run is executed until the model has been accepted. The screening study follows, and its analysis identifies the factors and interactions that change each response. The response surface study follows the screening analysis. The univariate assessment of elution flow rate may run in parallel with either designed study, since it shares no material or equipment constraint with them. Analysis, the proven acceptable range determination and the capability estimate are performed on the complete dataset, and the report is authored last.

No calendar dates are committed here. The schedule is held in PTP-001, which sequences this study against the other characterization studies of the campaign and against the Stage 2 activities that depend on them.

12 Risks and assumptions

Four risks to the study are recognized, and each carries a response.

The scale-down model may fail to represent the commercial column in a respect the qualification does not test. The qualification of §5.1 compares the attributes that enter and leave the step. It does not compare the flow distribution inside the two columns, which plate count and peak asymmetry bound only indirectly. Should a difference in packing quality or flow distribution be identified later, the affected conclusions would have to be revisited at scale.

The feed material may not be representative. The load for these studies is the neutralized pool of the low pH viral inactivation step, and its aggregate and host cell protein content set the starting point of every clearance the study measures. Load attributes will be measured on every run and reported with the results, so that a clearance can be distinguished from a low starting level. Where a lot proves out of trend on any load attribute, the runs performed on it should be identified in PCR-007 and their influence on the fitted models assessed.

The resin may behave differently at a different point in its life cycle. All characterization runs will be performed on resin within the cycle count qualified under SOP-2008, and the conclusions of this study are bounded by that cycle count. Performance at the end of the qualified life is established by the life cycle study and not here.

Assay variability may obscure a small effect. The centre point replicates give the reproducibility of each response under study conditions, and an effect smaller than that spread will not be claimed.

Three assumptions are made. The first is that the platform mechanisms apply to A-Mab at this step, which the model qualification and the centre point performance will confirm rather than assume. The second is that the parameters in Table 6.1 are the parameters that matter, which RA-001 established and which this study can test only within its own factor set. The third is that a quadratic model describes the responses adequately over the studied ranges, which the lack of fit test in §5.5 tests directly.

13 Approvals

This plan is approved by the functions recorded in Table 1. Execution may begin once the plan is approved and the scale-down model has been accepted under §5.1. Any change to the design, to the ranges or to the acceptance criteria after approval requires a revision of this plan carrying the same approvals, and the revision should be approved before the affected runs are executed.

14 Appendix A. Planned screening design matrix

The planned screening design, in the coded levels and the letter codes of Table 6.2. Response columns are absent because no run has been executed. The run numbers are labels rather than an execution sequence, since execution order will be randomized.

Table 14.1: Planned screening design matrix (coded levels; responses not yet measured).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

15 Appendix B. Planned response-surface design matrix

The planned response surface design, in the same coded levels and letter codes. The factorial runs repeat the corners of the screening design, the axial runs sit on the faces of the cube at the edge of each factor's studied range, and the centre point runs are executed at the set-point of every factor.

Table 15.1: Planned response-surface design matrix (coded levels; responses not yet measured).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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